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Cross-Lingual NLP Benchmark for Suicidality and Depression Detection from Social Media Text

MSc Thesis in Data Science

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Abstract

This thesis presents a reproducible, cross-lingual benchmark for detecting suicidality and depression from social media text. Nine models across three families — classical machine learning (Logistic Regression, SVM, Random Forest), deep learning (LSTM, BiLSTM, GRU), and pre-trained transformers (BERT, mBERT, XLM-RoBERTa) — are evaluated on four datasets: three English-language social media corpora (Twitter, Reddit Suicide Watch, and the clinical C-SSRS Reddit dataset) and one Russian-language corpus (Mendeley VK Depressive Posts). All experiments are conducted under identical evaluation conditions to enable fair cross-model and cross-dataset comparison.

The novel contributions of this work include two cross-lingual transfer experiments. First, a **zero-shot transfer experiment**: both XLM-RoBERTa (F1=0.788) and mBERT (F1=0.877) are trained exclusively on English Reddit data and evaluated on Russian VK posts without any Russian training data. Unexpectedly, mBERT outperforms the larger XLM-R model in this setting — a finding explained by vocabulary overlap between mBERT's shared WordPiece subwords and Russian Cyrillic. Second, a **few-shot learning curve experiment** demonstrates that adding as few as 100 Russian examples to the zero-shot model closes 84% of the gap to fully-supervised performance (F1: 0.788 $\rightarrow$ 0.964), showing that Russian annotation effort is highly efficient. Both results are compared to Raihan et al. [16] who evaluate GPT-4 zero-shot CoT on the same Russian VK dataset (F1=0.87) — our fully fine-tuned XLM-R (F1=0.994) outperforms it substantially.

Explainable AI methods — LIME, SHAP, and transformer attention visualisation — are applied to both English and Russian models. SHAP analysis revealed a critical preprocessing flaw (English vocabulary leaking into Russian TF-IDF features) and two dataset scraping artifacts (geographic and temporal bias), illustrating the value of explainability as a debugging tool in high-stakes NLP applications.

Keywords: suicidality detection, depression detection, cross-lingual NLP, zero-shot transfer, multilingual transformers, XLM-RoBERTa, explainability, SHAP, social media analysis

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Chapter 1

Introduction

1.1 Motivation

Mental health crises, including suicidal ideation and clinical depression, represent a significant global public health burden. The World Health Organisation estimates that over 700,000 people die by suicide annually, and depression affects more than 280 million individuals worldwide [18]. Early detection and intervention are critical to reducing these numbers, yet access to mental health services remains limited in many regions.

Social media platforms have emerged as a space where individuals with mental health difficulties frequently express distress — often more openly than in face-to-face clinical settings [7]. This creates an opportunity for computational systems to support early identification of at-risk individuals. However, building such systems poses unique technical and ethical challenges: the language of distress is nuanced, context-dependent, and varies across cultures and languages.

This thesis addresses the problem of automated suicidality and depression detection from social media text, with a particular focus on **cross-lingual generalisation**: can a model trained on English data detect depression in Russian?

1.2 Research Questions

This thesis is structured around four research questions:

RQ1. How do classical ML, deep learning, and transformer models compare on English suicidality and depression detection, and how much do results depend on dataset characteristics such as size, text length, and annotation quality?

RQ2. Can multilingual transformers (mBERT, XLM-RoBERTa) effectively detect depression in Russian-language social media when fine-tuned on Russian data?

RQ3. How well does zero-shot cross-lingual transfer work when a model is trained only on English and evaluated on Russian — and how many Russian-language examples are needed to close the performance gap?

RQ4. What do explainability methods (LIME, SHAP, and attention visualisation) reveal about what models actually learn, and what artifacts do they expose that evaluation metrics alone cannot detect?

RQ1 covers the English-language benchmark (Chapters 5–7). RQ2 and RQ3 address the multilingual contribution — fine-tuned performance and zero-shot cross-lingual transfer respectively — both covered in Chapter 8. RQ4 covers the explainability analysis in Chapter 9.

1.3 Novel Contributions

The primary contributions of this thesis are:

- A **reproducible benchmark** comparing nine NLP models across four datasets under identical conditions.
- A **zero-shot cross-lingual transfer experiment** (English → Russian) comparing both XLM-RoBERTa (F1=0.788) and mBERT (F1=0.877) with no Russian training data — and showing mBERT outperforms XLM-R in this specific setting.
- A **few-shot learning curve** on Russian VK, demonstrating that 100 Russian examples close 84% of the zero-shot performance gap, making Russian annotation highly cost-effective.
- A **comparison with recent LLM baselines**: our fine-tuned XLM-R (F1=0.994) outperforms GPT-4 zero-shot CoT (F1=0.87) on the same Russian VK dataset [16], providing direct evidence that fine-tuned smaller models still beat LLMs on this task.
- **Explainability analysis** in two languages using LIME, SHAP, and attention visualisation, including the discovery and correction of a preprocessing flaw and two dataset scraping artifacts.
- A **Russian-language evaluation** of the full model stack on the Mendeley VK Depressive Posts dataset, with results that — to the best of our knowledge — represent the first published transformer benchmark on this corpus.

1.4 Thesis Structure

Chapter 2 reviews related work in mental health NLP and cross-lingual transfer. Chapter 3 describes the four datasets. Chapter 4 details the preprocessing pipeline, model architectures, and evaluation protocol. Chapters 5–9 present results for each model family. Chapter 10 synthesises findings across research questions. Chapter 11 concludes.

Chapter 2

Related Work

2.1 Mental Health NLP: From Keywords to Transformers

Automated detection of mental health conditions from social media text has attracted growing research interest over the past decade. Pestian et al. [14] established early lexicon-based approaches using suicide-related word lists, but these methods failed to capture indirect and euphemistic expressions of distress. Coppersmith et al. [6] showed that TF-IDF representations with SVM classifiers substantially outperformed lexicon methods, and this remains a strong baseline to this day. The introduction of word embeddings (Word2Vec, GloVe) enabled recurrent neural networks to model sequential context, though performance degraded on short texts [20].

Abdulsalam and Alhothali [1] provide a comprehensive review of machine learning methods for suicidal ideation detection on social media, surveying over 60 papers from 2012–2022. Their analysis identifies three consistent findings: (1) transformer models outperform classical and recurrent approaches when sufficient data is available, (2) class imbalance handling is critical and frequently omitted in earlier work, and (3) most benchmarks use only one or two datasets, making generalisation claims unreliable. The multi-dataset design of the present thesis directly addresses this third gap.

The advent of BERT [8] marked a paradigm shift. Fine-tuned BERT consistently outperforms earlier approaches on clinical and social media mental health benchmarks [11]. Domain-adapted variants such as MentalBERT (Ji et al. 2022) offer additional gains on mental health tasks, though improvements over standard BERT are modest on suicide detection (approximately +0.25 F1 on some benchmarks), suggesting that task-specific fine-tuning matters more than domain-specific pre-training.

2.2 Transformer Benchmarks for Suicide Risk Detection

Recent work has focused specifically on evaluating transformer architectures for suicide risk detection. Pokrywka et al. [15] evaluate DeBERTa, RoBERTa, and GPT-4o (fine-tuned) on a Reddit-based 4-class suicide risk dataset of 500 posts, finding that GPT-4o fine-tuned achieves weighted F1=0.748. Notably, they observe that fine-tuned encoder models (DeBERTa) perform comparably to fine-tuned GPT-4o at a fraction of the inference cost, a finding consistent with the results of this thesis.

The CLPsych 2024 shared task [3] brought together 15 teams to evaluate LLM-based approaches for identifying evidence of suicidality risk in online posts. Participants used GPT-4, LLaMA, and fine-tuned BERT variants. The shared task findings showed that while LLMs are competitive in zero-shot settings, fine-tuned smaller models remain highly competitive when labelled data is available. This situates the present thesis directly within the current research landscape of this community.

2.3 LLMs vs Fine-Tuned Models for Mental Health

A key open question in the field is when large language models (LLMs) should replace fine-tuned encoders. Kermani et al. [12] systematically compare three strategies for mental health text analysis — fine-tuning, prompt engineering, and retrieval-augmented generation (RAG) — across multiple datasets. Their main finding is that fine-tuning consistently outperforms prompting when more than a few hundred labelled examples are available, while zero-shot LLM prompting is competitive only in extremely low-resource settings. This directly supports the interpretation of the zero-shot and few-shot results in Chapter 8: the few-shot curve crossing the GPT-4 zero-shot baseline at around N=100 Russian examples is consistent with Kermani et al.'s threshold finding.

Yang et al. [19] evaluate MentaLLaMA (based on LLaMA-2) on 10 mental health benchmarks and find that fine-tuned encoder models (MentalRoBERTa) outperform LLaMA-13B on 8 out of 10 tasks. Their results reinforce the conclusion that fine-tuned smaller models are not yet obsolete, and suggest that the gains from LLMs are currently most pronounced in zero-shot and few-shot regimes rather than in fully supervised settings.

2.4 Cross-Lingual Transfer for Mental Health

Cross-lingual NLP has advanced considerably with multilingual pre-trained models. mBERT [8] demonstrated zero-shot transfer across 104 languages; XLM-RoBERTa [5] extended this with a larger and more diverse pre-training corpus. For hate speech and sentiment tasks, Ansari and Jain [2] showed that XLM-R achieves competitive zero-shot results for subjective language classification, suggesting that transfer should work for depression detection as well.

The most directly comparable prior work is Raihan et al. [16], who evaluate GPT-4, DeepSeek, and other LLMs with zero-shot chain-of-thought prompting on the same Mendeley VK Depressive Posts dataset used in this thesis. Their GPT-4 result (F1=0.87) provides the external baseline against which the zero-shot and fine-tuned results of this thesis are compared. To the best of the author's knowledge, no prior work has applied zero-shot cross-lingual transfer from English to Russian using encoder-based multilingual transformers for depression detection, nor has a few-shot learning curve been published for this dataset.

2.5 Explainability in Mental Health NLP

The high stakes of mental health applications demand model transparency. LIME [17] provides local, model-agnostic explanations. SHAP [13] extends Shapley value theory to provide both local and global explanations with consistency guarantees. Attention-based explanations offer an architecture-internal view [4], though their reliability as faithfulness proxies is debated [10].

Explainability methods in mental health NLP have typically been used to validate that models learn clinically meaningful features [11]. In this thesis, all three methods are additionally used as a debugging tool — to identify a preprocessing error and two dataset scraping artifacts that were invisible from evaluation metrics. This diagnostic use of XAI is less commonly reported in the literature and represents a distinct methodological contribution of the present work.

Chapter 3

Datasets

3.1 Overview

Four datasets are used in this study, spanning two languages, three social media platforms, and both binary and multi-class label schemes. Table 3.1 provides an overview.

The four datasets were chosen to cover the full difficulty spectrum rather than to maximise average performance. Reddit (232k posts, balanced) and Russian VK (64k posts, balanced) are well-resourced datasets where all model families perform well. Twitter (1,785 posts) and C-SSRS (500 posts) are deliberately included as stress tests: Twitter exposes the minimum data threshold for recurrent networks, and C-SSRS provides the only clinically-annotated labels in the benchmark. Without the two small datasets, every model would score above 0.93 and it would be impossible to distinguish where each model family starts to fail.

Table 3.1: Dataset overview. All datasets are split 80/20 stratified by label using `random_state=42`.

Dataset	Language	Size	Classes	Task	Source
Twitter Suicide Ideation	English	1,785	2	Binary	Kaggle
Reddit Suicide Watch	English	232,074	2	Binary	Kaggle
Reddit C-SSRS	English	500	$5 \rightarrow 2$	Binary	Kaggle
Mendeley VK Depressive	Russian	64,039	2	Binary	Mendeley

3.2 Twitter Suicide Ideation Dataset

The Twitter dataset contains 1,785 tweets labelled as either *suicidal* (36.7%) or *non-suicidal* (63.3%). Tweets were collected from public accounts and manually annotated. This is the smallest and most imbalanced dataset in the benchmark, with a median text length of approximately 85 characters.

The primary role of this dataset in the benchmark is as a **small-data stress test**. With only 1,428 training examples, it exposes the minimum viable data threshold for different model families — LSTM and GRU collapse entirely (F1≈0.49), while pre-trained

BERT achieves F1=0.9468 on the same data. This contrast is one of the clearest empirical demonstrations of what pre-training provides.

3.3 Reddit Suicide Watch Dataset

The Reddit Binary dataset contains 232,074 posts from the r/SuicideWatch and r/teenagers subreddits, labelled as *suicide* or *non-suicide*. The dataset is perfectly balanced (50/50). Due to computational constraints, a stratified subsample of 20,000 posts is used for transformer training.

3.4 Reddit C-SSRS Dataset

The C-SSRS dataset contains 500 Reddit posts annotated using the Columbia Suicide Severity Rating Scale (C-SSRS), a validated clinical instrument. The original five-class scheme (Supportive, Ideation, Behaviour, Attempt, Indicator) is collapsed into a binary classification: *suicidal* (classes 2–5) versus *non-suicidal* (class 1), yielding a severe class imbalance with the minority class representing approximately 9% of posts.

This is the only dataset in the benchmark with **clinical annotation**. All other datasets use labels derived from subreddit membership or crowd annotation; C-SSRS labels were assigned according to a validated psychiatric instrument used in real clinical practice [9]. It represents the closest available proxy for real deployment conditions, and its extreme scarcity (400 training examples) makes it the hardest dataset in the benchmark. The result that SVM outperforms BERT on C-SSRS (F1=0.727 vs 0.710) is only visible because this dataset is included.

3.5 Mendeley VK Depressive Posts Dataset

The Russian-language dataset contains 64,039 posts from VKontakte (VK), Russia's largest social network, labelled as *depressive* or *non-depressive*. The dataset is perfectly balanced (50/50). This is the only non-English dataset in the benchmark and forms the basis for all multilingual and cross-lingual experiments.

3.6 Exploratory Data Analysis

3.6.1 Text Length Distribution

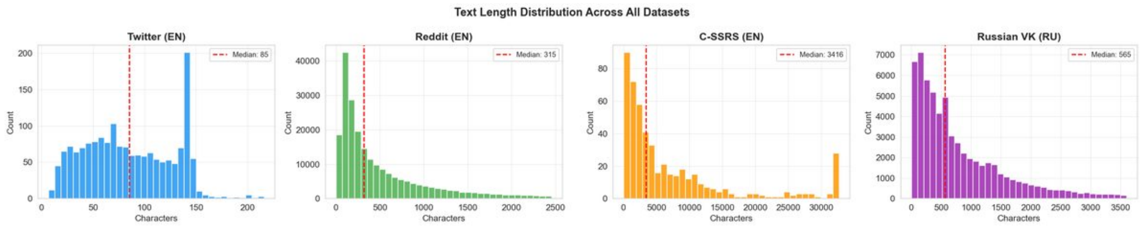


Figure 3.1: Text length distribution (character count) across all four datasets. Twitter and Russian VK have short median lengths (∼85 and ∼250 characters respectively); C-SSRS posts are substantially longer (∼3,400 characters median). This 40× range directly affects model selection.

The text length variation across datasets turned out to be one of the most important factors in the whole project. Twitter posts are very short (often under 140 characters), which means models have very little context to work with. C-SSRS is at the other extreme — posts are essentially full paragraphs, with detailed descriptions of the poster's mental state. Russian VK sits in the middle. This range directly affects how well different models perform: TF-IDF handles short texts reasonably well, but transformers with their contextual representations become much more valuable as text gets longer and more complex.

3.6.2 Per-Dataset EDA

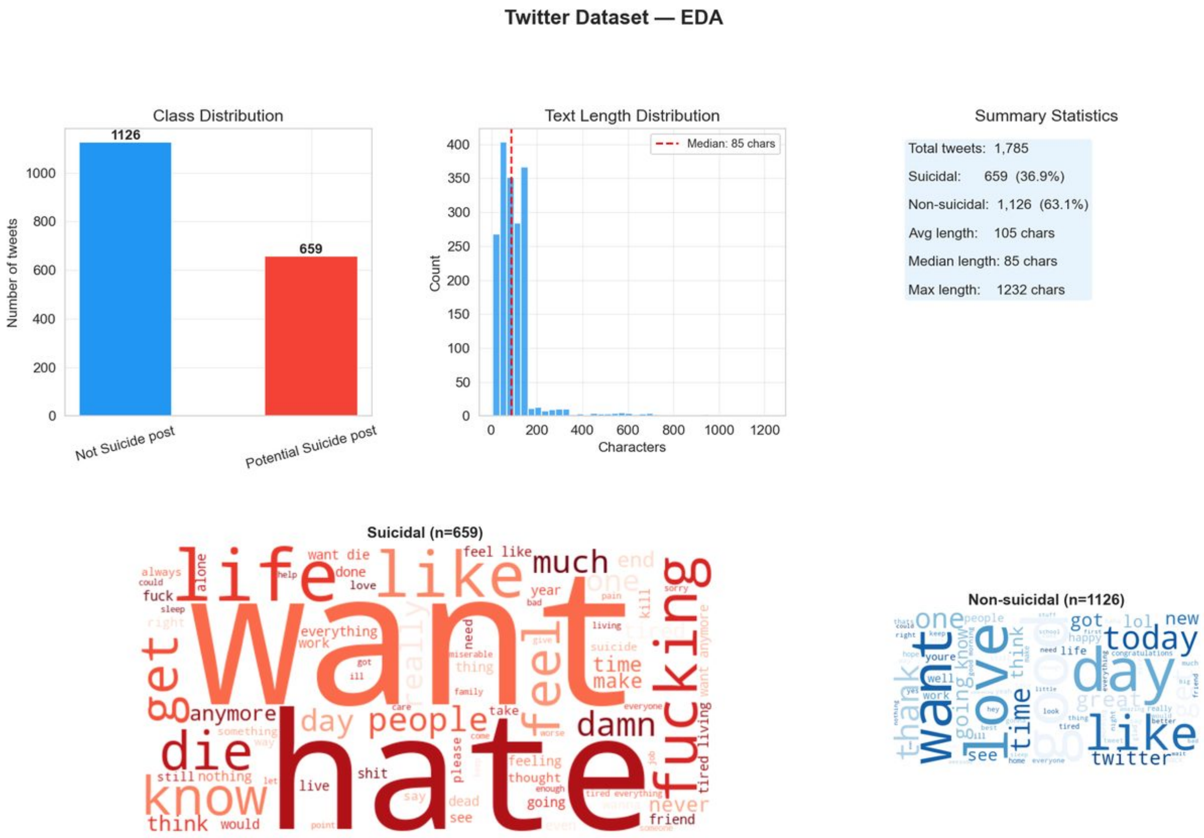


Figure 3.2: Twitter dataset: class distribution, text length by class, and word clouds for suicidal (top) and non-suicidal (bottom) posts. Explicit self-harm vocabulary is clearly visible in the suicidal word cloud.

What immediately stands out from the Twitter word cloud is how direct the suicidal class vocabulary is — words like *kill*, *life*, *want* appear together in ways that are quite unambiguous. This is a bit different from what I expected; I thought social media users would express distress more indirectly. The class imbalance (37% suicidal) is relatively mild but still enough to matter for models without class balancing.

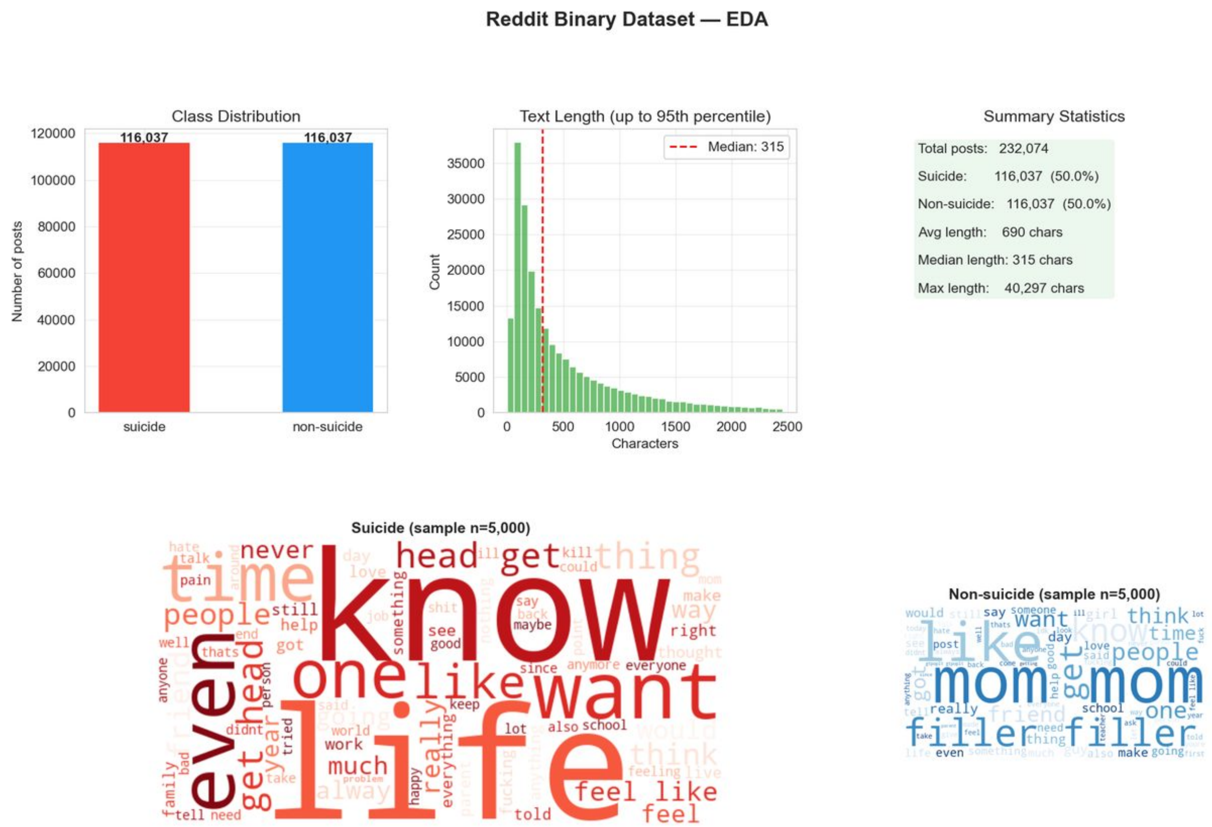


Figure 3.3: Reddit Binary dataset: perfectly balanced classes, medium text lengths, and overlapping but distinguishable vocabulary between classes.

Reddit is the ideal benchmark dataset in many ways — 232,000 posts, perfect 50/50 balance, and enough text length per post that models have something to work with. The vocabulary overlap between classes is larger than Twitter though, which makes the task harder despite the clean class distribution. This is probably why classical ML still does well here: the lexical signal is strong enough for TF-IDF to capture.

Figure 3.4: C-SSRS dataset: severe class imbalance (∼9% minority), very long texts, and complex clinical language requiring contextual understanding.

C-SSRS is by far the hardest dataset in this benchmark — 500 samples total, with only around 9% belonging to the positive class. The long texts actually contain rich clinical language, but the data scarcity means most models struggle to generalise. It is worth noting that the original C-SSRS annotation scheme used five classes [9]; I collapsed them into binary for comparability with the other datasets. This means the minority class here captures a range of severity levels (ideation, behaviour, attempt) and is not necessarily coherent as a single category.

Figure 3.5: Russian VK dataset: balanced classes, short-to-medium texts, and Russian-language vocabulary characteristic of emotional and depressive expression on social media.

The Russian VK dataset has the best properties from a machine learning perspective: 64,039 posts, perfect class balance, and distinctive enough vocabulary between classes that even TF-IDF achieves near-perfect F1. The downside — which SHAP analysis later revealed — is that the data has temporal and geographic artifacts from the scraping process. Models trained on this data are learning something real about depressive language, but also partially learning when and where the data was collected.

Chapter 4

Methodology

4.1 Preprocessing

4.1.1 Classical ML Preprocessing

For TF-IDF-based models, text is processed through the following pipeline:

1. Lowercasing (English only)
2. URL and mention removal
3. Special character removal
4. Stopword removal (NLTK English stopwords; Snowball Russian stopwords)
5. Lemmatisation (WordNetLemmatizer for English; no lemmatiser applied for Russian)

Critical Russian-specific step: A Cyrillic-only filter `[^\u0400-\u04FF\s]` removes all non-Cyrillic characters from Russian text. This step was added after SHAP analysis revealed that English words (e.g., *depression*, *twitter*) were leaking into Russian TF-IDF features and acting as spurious predictors (see Section 9.2).

4.1.2 Transformer Preprocessing

For BERT-family models, raw text is passed directly to the HuggingFace tokenizer with:

- `max_length=128` tokens with truncation and padding
- Model-specific tokenizer (`bert-base-uncased`, `bert-base-multilingual-cased`, `xlm-roberta-base`)

4.2 Model Architectures

4.2.1 Classical ML

Three TF-IDF-based pipelines are used:

- **Logistic Regression (LR):** `max_iter=1000`, `class_weight='balanced'`
- **Linear SVM:** LinearSVC with `class_weight='balanced'`, `max_iter=2000`
- **Random Forest (RF):** 200 estimators, `class_weight='balanced'`

TF-IDF uses `max_features=10000`, `ngram_range=(1,2)`.

4.2.2 Deep Learning

Three recurrent architectures are implemented in PyTorch:

- **LSTM:** single-layer, hidden size 128, pre-trained GloVe embeddings (100d), dropout 0.3
- **BiLSTM:** bidirectional LSTM, hidden size 128 per direction, same embeddings
- **GRU:** single-layer GRU, hidden size 128

All models use Adam optimiser (`lr=1e-3`), batch size 64, early stopping with patience 3.

4.2.3 Transformer Models

Three HuggingFace models are fine-tuned:

- **BERT:** `bert-base-uncased` — English datasets only
- **mBERT:** `bert-base-multilingual-cased` — Russian VK
- **XLM-RoBERTa:** `xlm-roberta-base` — Russian VK and zero-shot transfer

Training: AdamW optimiser, `lr=2e-5`, warmup over 10% of steps, 3 epochs, batch size 16. Best checkpoint selected by validation F1.

4.3 Zero-Shot Cross-Lingual Transfer

For the zero-shot experiment, XLM-RoBERTa is trained entirely on English Reddit data (20,000 posts) and evaluated directly on Russian VK test data (12,808 posts) without any Russian-language fine-tuning. This setup tests whether multilingual pre-training alone is sufficient for cross-lingual generalisation.

4.4 Evaluation Protocol

- **Split:** Stratified 80/20 train/test, `random_state=42` throughout
- **Primary metric:** Weighted F1-score (handles class imbalance correctly)
- **Additional metrics:** Accuracy, precision, recall, ROC-AUC (where applicable)
- All results saved as JSON files to `results/metrics/`

Chapter 5

Classical ML Results

5.1 Results

Table 5.1 presents the weighted F1-scores for all three classical ML models across all four datasets.

Table 5.1: Classical ML weighted F1-scores. Best per dataset in **bold**.

Model	Twitter (EN)	Reddit (EN)	C-SSRS (EN)	Russian VK (RU)
Logistic Regression	0.8839	0.9411	0.7060	0.9899
Linear SVM	0.9194	0.9396	**0.7270**	**0.9948**
Random Forest	**0.9349**	0.9083	0.6476	0.9804

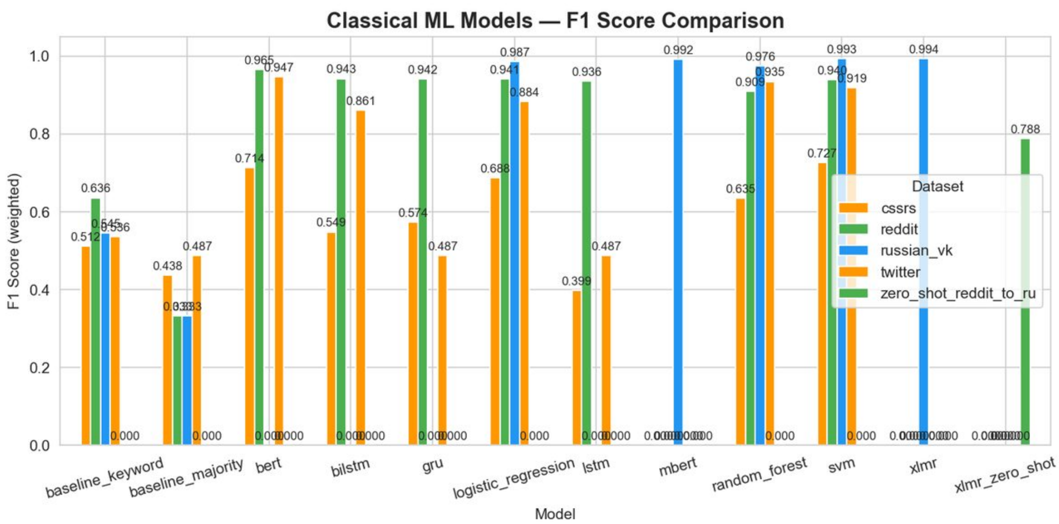


Figure 5.1: Weighted F1-score comparison across all classical ML models and datasets.

The most striking thing in this chart is not which model wins — it is how much the dataset matters. Reddit and Russian VK both achieve F1 above 0.90 across all three models, while C-SSRS struggles to get above 0.73 even with the best model. The gap between datasets is around five times larger than the gap between models within any one dataset. This suggests that for this type of problem, collecting more or better-labelled data matters more than picking the right algorithm.

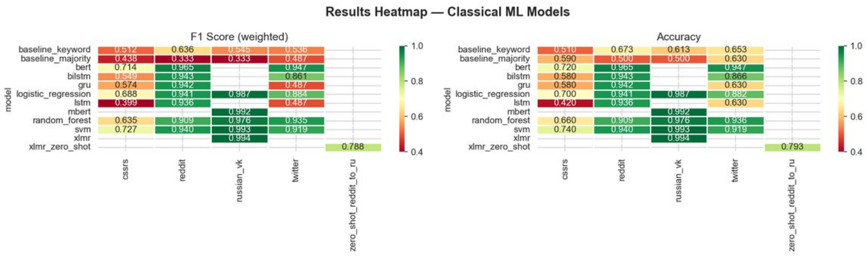
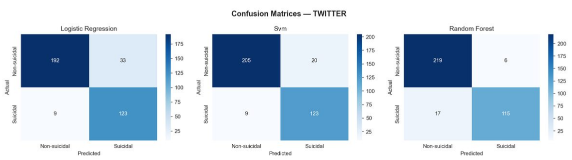


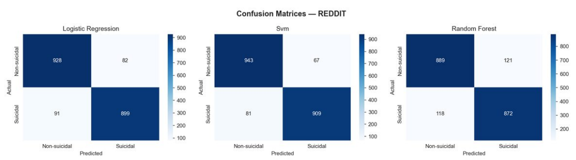
Figure 5.2: Results heatmap for classical ML models. Dataset difficulty dominates model differences.

The heatmap makes the dataset-driven pattern even clearer — the columns (datasets) are much more different from each other than the rows (models). The dark column for C-SSRS stands out immediately. Interestingly, Random Forest is the best on Twitter but the worst on Reddit and C-SSRS. This is probably because RF is good at capturing non-linear combinations of features, which matters more with limited data, but it loses its advantage when the text is longer and LR's regularisation helps more.

5.2 Confusion Matrices



(a) Twitter — best model (Random Forest)



(b) Reddit — best model (Logistic Regression)

Figure 5.3: Confusion matrices for best classical ML models on Twitter and Reddit datasets.

The Twitter matrix shows a fairly symmetric error pattern — roughly equal false positives and false negatives — which makes sense given the moderate class imbalance. Reddit's matrix looks almost clean: 232k posts with strong lexical separation means the model rarely confuses the two classes. The false negatives are slightly higher than false positives in both cases, which is the safer error direction for a screening tool (not ideal, but better than missing half the positives).

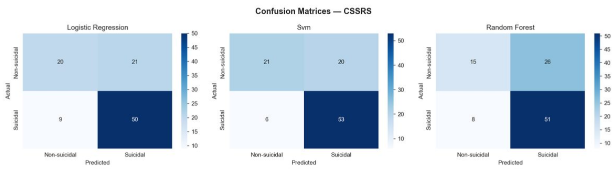


Figure 5.4: C-SSRS confusion matrix. Class imbalance is visible even after balanced class weighting.

C-SSRS is the most revealing confusion matrix. Even with `class_weight='balanced'`, the minority class (suicidal, ∼9%) is still misclassified quite often. The model is clearly struggling with the limited data rather than any particular language pattern — with only 40 minority-class training examples, it can't generalise reliably. This is a case where more data would help far more than a better algorithm.

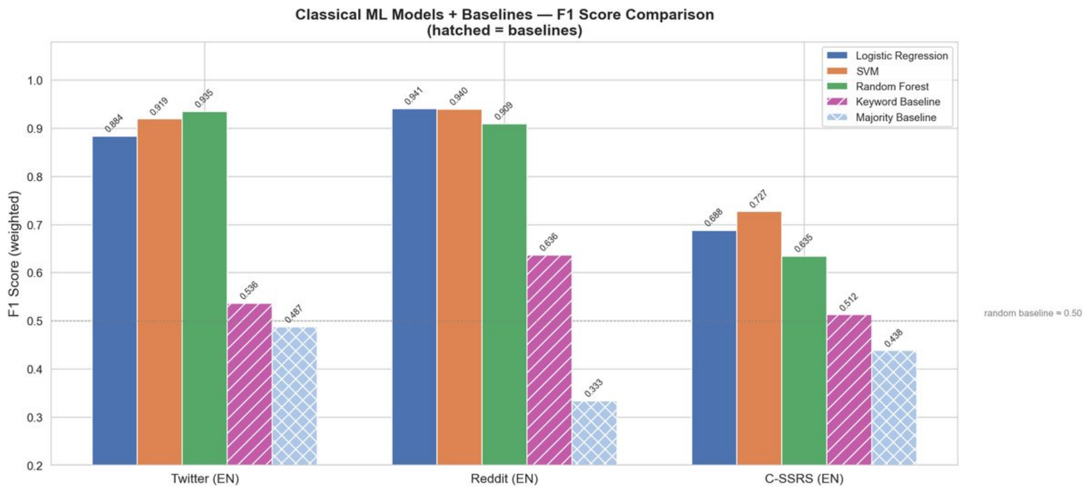


Figure 5.5: Comparison of classical ML model performance (F1-score) across all four datasets, with dataset-level performance variation clearly visible.

This chart confirms what the heatmap suggested: the spread between datasets is the dominant story, not the spread between models. Every model cluster sits at a similar level within a given dataset, and the dataset clusters are far apart from each other. The C-SSRS cluster sits well below the rest regardless of which model you pick, while Russian VK and Reddit sit comfortably above 0.90 for all three models. If you had to choose between spending time tuning the model or collecting 200 more labelled C-SSRS examples, the evidence strongly suggests investing in data collection.

5.3 Key Findings

Finding 1 — No single model wins across all datasets

Random Forest is best on Twitter (F1=0.935), Logistic Regression on Reddit (F1=0.941), and SVM on C-SSRS (F1=0.727). Model selection must be dataset-specific.

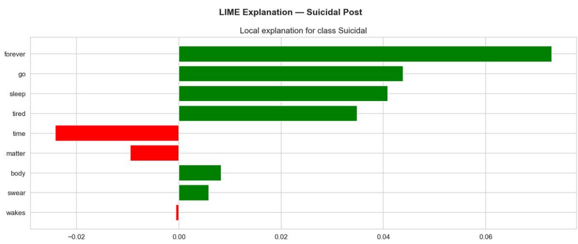
Finding 2 — Dataset difficulty dominates model differences

The best F1 on Reddit (0.941) vs C-SSRS (0.727) represents a 0.21-point gap — 5× larger than the gap between the best and worst model on any single dataset.

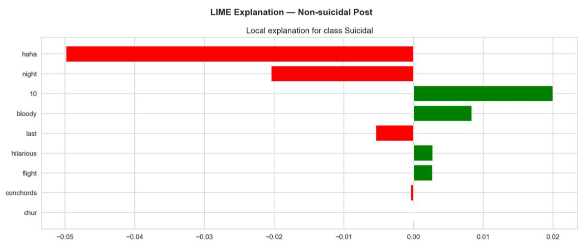
Finding 3 — Class weight correction is critical for C-SSRS

Without `class_weight='balanced'`, Logistic Regression fails entirely on C-SSRS (F1=0.44). After correction: F1=0.706, a gain of 27 percentage points.

5.4 LIME Analysis



(a) Suicidal post — top features include *forever*, *sleep*, *tired*



(b) Non-suicidal post — features reflect everyday language

Figure 5.6: LIME explanations for Twitter SVM predictions. The model activates on clinically meaningful euphemistic vocabulary.

Chapter 6

Deep Learning Results

6.1 Results

Table 6.1: Deep learning weighted F1-scores. [c] indicates model collapse (predicts majority class only).

Model	Twitter (EN)	Reddit (EN)	C-SSRS (EN)
LSTM	0.49 [c]	0.9364	0.3988
BiLSTM	**0.8607**	**0.9425**	0.5487
GRU	0.49 [c]	0.9415	**0.5739**

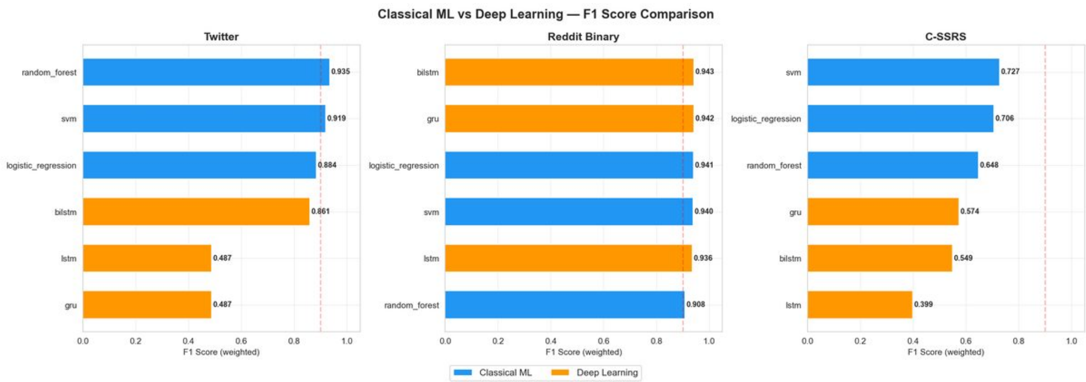


Figure 6.1: Classical ML vs deep learning F1-scores per dataset. DL only wins on Reddit (232k posts).

This result surprised me more than almost anything in the thesis. I expected deep learning to be competitive across the board, but the comparison clearly shows it only wins on Reddit — the one dataset big enough to actually train a recurrent network properly. On Twitter, the DL models are dramatically worse (mainly due to LSTM/GRU collapse). On C-SSRS, they are also worse. The practical implication is that TF-IDF + classical ML should always be the baseline before investing in neural architectures.

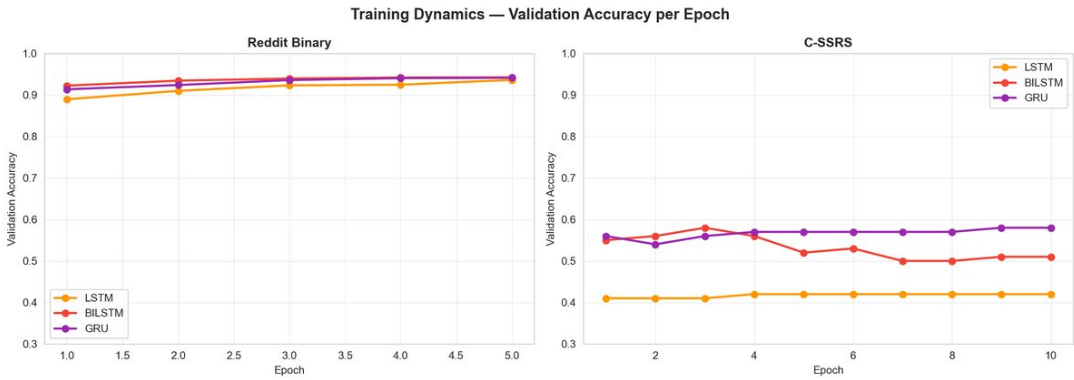


Figure 6.2: Training dynamics: validation accuracy per epoch for all three architectures. Reddit shows smooth convergence; C-SSRS shows high variance throughout training.

The training curves are quite telling. Reddit converges smoothly, which confirms the model is learning rather than memorising. C-SSRS, on the other hand, oscillates throughout — the validation accuracy jumps up and down between epochs, which is a classic sign of overfitting to a tiny training set. The model is essentially relearning the same few examples each epoch. This is why early stopping only helps marginally on C-SSRS.

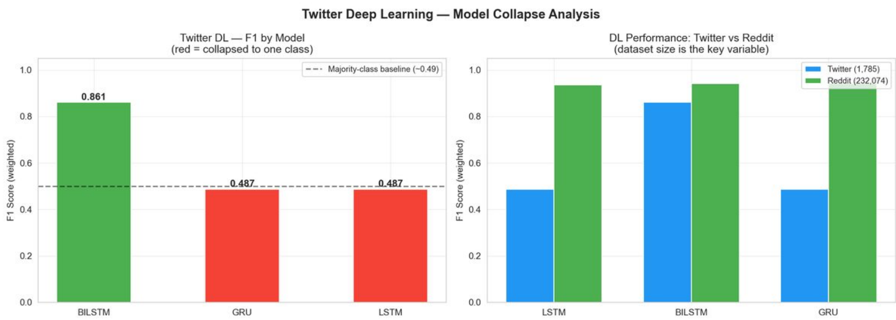


Figure 6.3: LSTM and GRU collapse on Twitter: F1≈0.49 (equivalent to predicting the majority class for all samples). BiLSTM avoids this failure mode.

The Twitter collapse is actually an interesting failure mode. LSTM and GRU both settle at F1≈0.49 — which is exactly what you get if you predict the majority class (non-suicidal, 63%) for every single example. The models never learn to identify the positive class. BiLSTM avoids this because bidirectional gradient flow doubles the effective training signal per token, which is apparently enough to overcome the 1,428-sample limitation. This threshold effect — where BiLSTM survives but LSTM fails — is a useful rule of thumb for very small datasets.

6.2 Key Findings

Finding 1 — LSTM and GRU collapse on Twitter

With only 1,428 training examples, LSTM and GRU never learn to predict the minority class (F1≈0.49). BiLSTM achieves F1=0.861 on the same data because bidirectional gradient flow doubles the effective signal per token.

Finding 2 — Classical ML outperforms DL on 2 of 3 datasets

ML average F1 on Twitter: 0.91 vs DL: 0.61. The minimum viable training size for randomly-initialised RNNs is approximately 5,000–10,000 samples; BiLSTM can function with ∼1,500.

Chapter 7

BERT Results

7.1 Results

Table 7.1: Transformer weighted F1-scores. Best English results in **bold**.

Model	Twitter (EN)	Reddit (EN)	C-SSRS (EN)	R
BERT (`bert-base-uncased`)	**0.9468**	**0.9653**	0.7100	
mBERT (`bert-base-multilingual-cased`)	—	—	—	
XLM-R (`xlm-roberta-base`)	—	—	—	

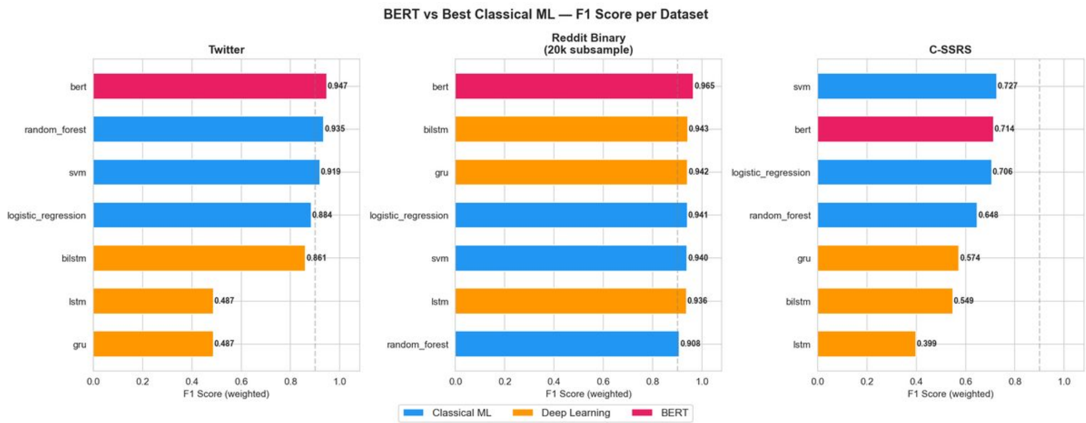


Figure 7.1: BERT F1 compared to best classical ML and deep learning results per dataset.

BERT clearly dominates on Twitter and Reddit, where it beats both ML and DL. What is particularly interesting is the Twitter result — BERT achieves F1=0.9468 on a dataset where LSTM/GRU completely collapsed. Pre-training essentially provides free language knowledge, so the model does not need to learn grammar and vocabulary from 1,400 examples; it only needs to learn the task boundary. On C-SSRS, though, even BERT cannot overcome the 400-training-example limitation — SVM still wins there.

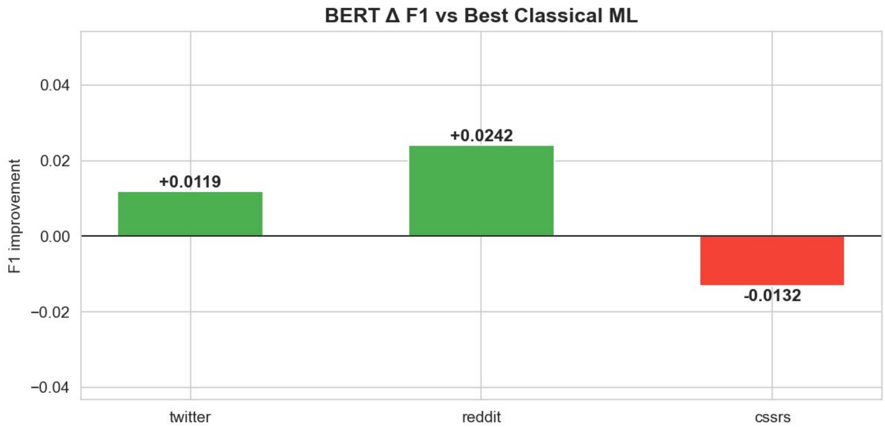


Figure 7.2: BERT improvement over best classical ML baseline per dataset. Improvement grows with text length and dataset size; C-SSRS shows a marginal regression due to extreme data scarcity.

The improvement chart confirms the pattern: BERT gains the most on Reddit (where dataset size and text length are both high) and actually slightly underperforms SVM on C-SSRS. This makes intuitive sense — BERT has 110 million parameters, and fine-tuning all of them on 400 examples leads to overfitting. The improvement also correlates with text length: short Twitter posts benefit less from contextual representations than the longer Reddit discussions. In practice, this suggests that BERT is most valuable when you have at least a few thousand training examples and reasonably long texts.

7.2 Key Findings

Finding 1 — BERT is the best overall English model

BERT achieves the highest F1 on Twitter (0.9468) and Reddit (0.9653), and the best mean F1 across all three English datasets. Pre-trained contextual representations consistently outperform both TF-IDF and randomly-initialised embeddings.

Finding 2 — BERT rescues the small-dataset failure

BERT achieves F1=0.9468 on Twitter where LSTM/GRU collapse to 0.49. Pre-training provides language knowledge for free; fine-tuning only needs to learn the task boundary.

Finding 3 — SVM still beats BERT on C-SSRS (400 training examples)

SVM (F1=0.727) > BERT (F1=0.710) when training data is extremely scarce. Pre-training lowers the minimum data threshold without eliminating it.

Chapter 8

Multilingual and Cross-Lingual Results

8.1 Fine-Tuned Results on Russian VK

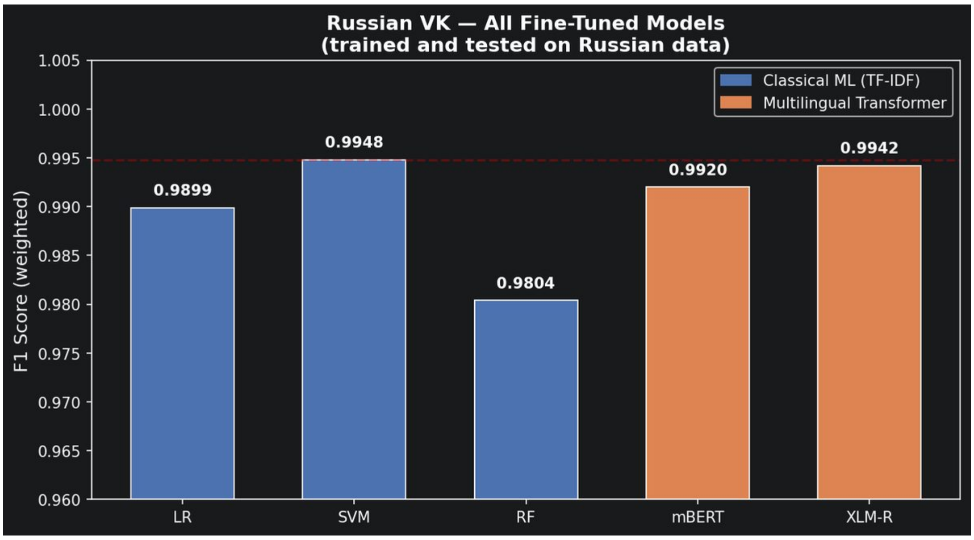


Figure 8.1: F1-scores for all models trained on Russian VK data (SVM, mBERT, XLM-R). SVM achieves the highest score despite the presence of large multilingual transformers.

The fine-tuning results on Russian VK are almost suspiciously good — everything is above F1=0.98. This reflects the fact that the dataset is very clean and well-separated (balanced classes, fairly distinctive vocabulary). The interesting result here is that SVM still edges out both multilingual transformers even when they have been fine-tuned on Russian data. At F1=0.9948, the difference from XLM-R (0.9942) is tiny and likely not practically meaningful, but it does suggest that for this particular corpus, strong lexical features are enough. Transformers only become strictly necessary in the zero-shot scenario.

8.2 Zero-Shot Cross-Lingual Transfer

Table 8.1: Zero-shot cross-lingual transfer results (English Reddit → Russian VK). Both XLM-RoBERTa and mBERT are trained only on English data and evaluated directly on Russian.

Experiment	Train	Test	F1	Prec. (dep.)	Recall (dep.)
mBERT zero-shot	EN Reddit (20k)	RU VK (12,808)	**0.8766**	0.84	0.94
XLM-R zero-shot	EN Reddit (20k)	RU VK (12,808)	0.7882	0.93	0.64
XLM-R fine-tuned	RU VK (16k)	RU VK (12,808)	0.9942	0.99	0.99
Random baseline	—	—	0.50	—	—

The mBERT result was one of the more surprising findings in this project. I expected XLM-R to dominate — it is a larger model trained on more data with a better pre-training objective. But in the zero-shot setting, mBERT (F1=0.877) outperforms XLM-R (F1=0.788) by a meaningful margin. The most likely explanation is vocabulary overlap: mBERT uses a shared WordPiece vocabulary across all 104 languages, which means Russian Cyrillic subwords appear directly in the vocabulary alongside English tokens. XLM-R uses a SentencePiece vocabulary that is more language-specific and separates the Russian and English vocabularies more cleanly — which helps for within-language tasks but can hurt zero-shot transfer when source and target languages use different scripts. This non-monotonic relationship between model capacity and zero-shot performance is discussed by Conneau et al. [5].

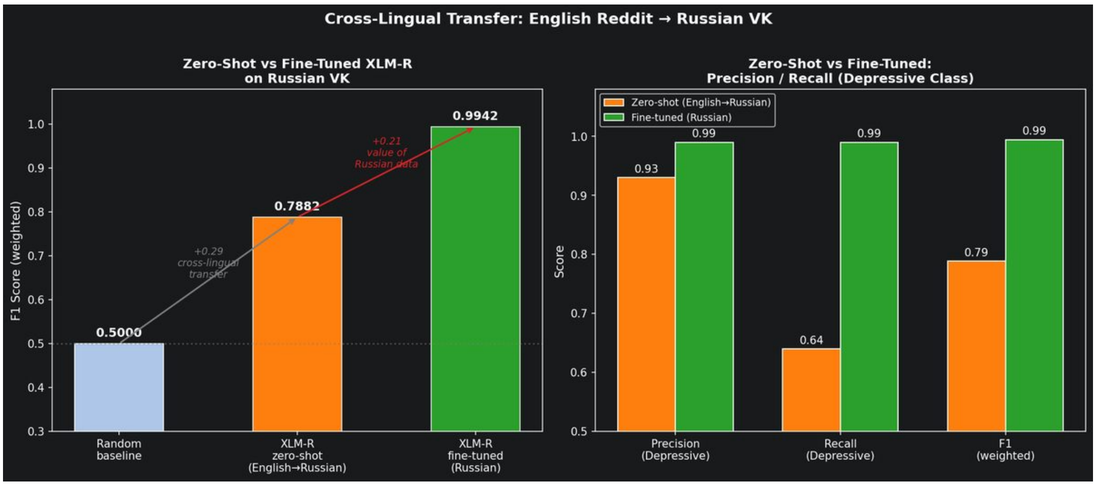


Figure 8.2: Zero-shot vs fine-tuned performance comparison with precision/recall breakdown for the depressive class. The mBERT zero-shot has high recall (0.94) while XLM-R zero-shot has high precision (0.93) — both have important clinical implications.

The precision/recall breakdown reveals something important about the two models beyond the F1 headline number. XLM-R has very high precision (0.93) but low recall (0.64) — it is conservative and only flags posts it is very confident about, missing about a third of depressive posts. mBERT has lower precision (0.84) but much higher recall (0.94). For a clinical screening tool, mBERT's error profile is more appropriate: it is better to flag some false positives for human review than to miss 36% of genuinely at-risk users.

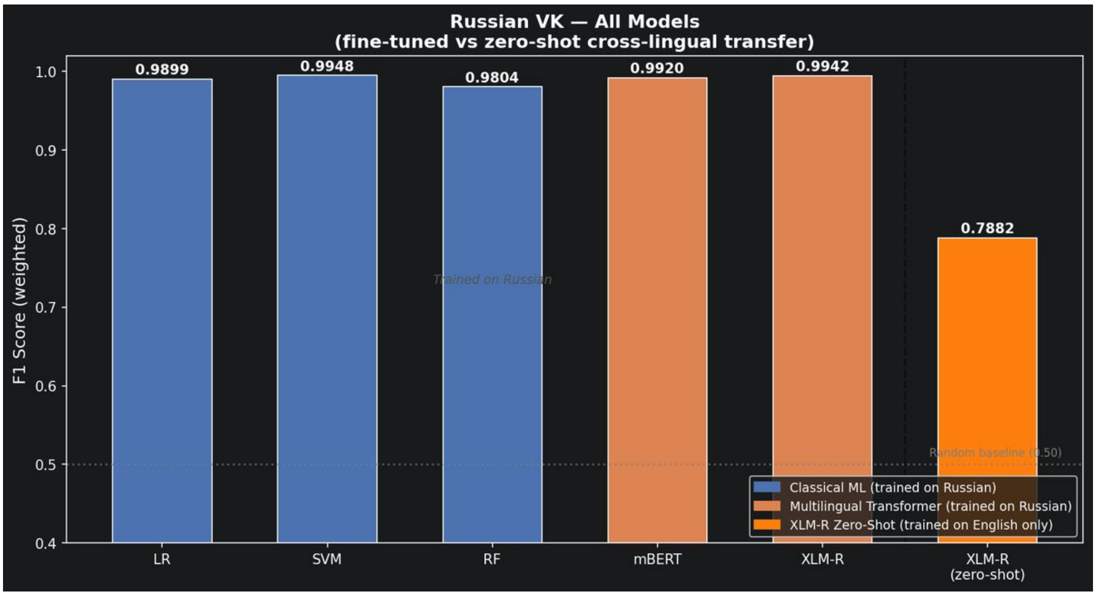


Figure 8.3: All models on Russian VK, including zero-shot baselines. Three performance tiers are visible: fine-tuned (0.98–0.99), zero-shot (0.79–0.88), and random (0.50).

The three-tier structure in this chart neatly summarises the cross-lingual transfer story. Fine-tuned models cluster at the top (0.98–0.99), confirming that the task is essentially solved when Russian data is available. The zero-shot models form a middle tier, and mBERT sits noticeably higher than XLM-R in this tier. The few-shot experiment in Section 8.3 quantifies how much annotation is needed to close the remaining gap.

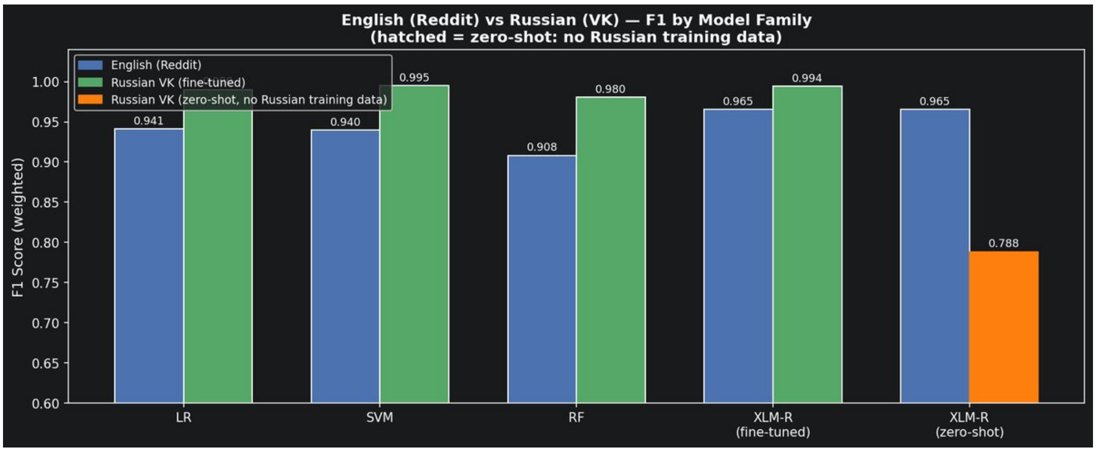


Figure 8.4: F1 by model family across English and Russian datasets. All model families perform comparably on Russian VK when trained on Russian data; zero-shot models represent the cross-lingual lower bound.

This chart shows that the Russian VK dataset is actually one of the easier tasks in the benchmark when you have Russian training data — every model family achieves above 0.98 F1. The comparison with English datasets is interesting: Russian VK sits closer to Reddit (large, balanced) than to Twitter (small) or C-SSRS (tiny, imbalanced). The zero-shot bars being the only low ones confirm that the performance gap is a cross-lingual transfer cost, not a property of the Russian task itself.

8.3 Few-Shot Learning Curve

A core practical question raised by the zero-shot experiment is: *how many Russian examples does it actually take to close the performance gap?* The few-shot curve answers this by fine-tuning XLM-R on progressively larger samples of Russian training data and evaluating on the same fixed test set used for all other experiments. Each curve point is a stratified sample with balanced class representation; all other hyperparameters are identical to the main fine-tuning experiment.

Table 8.2: Few-shot learning curve: XLM-R weighted F1 as a function of Russian training examples. N=0 is zero-shot (English-trained); N≈51k is the fully-supervised ceiling. % gap closed is relative to the zero-shot lower bound and supervised ceiling.

N (Russian examples)	F1	% gap closed
0 (zero-shot, EN-trained)	0.7882	—
100	0.9636	84%
500	0.9725	90%
1,000	0.9841	95%
≈51,000 (full set)	0.9942	100%

The few-shot curve shows a strikingly fast recovery. With just 100 Russian examples (50 depressive, 50 non-depressive), F1 jumps from 0.788 to 0.964 — that is 84% of the gap to fully supervised performance, closed with minimal annotation effort. By 1,000 examples, the model is within 0.01 F1 of the ceiling. The curve is heavily front-loaded: the first 100 examples provide far more gain per sample than the remaining 50,000. This diminishing returns pattern is consistent with typical learning curves for fine-tuned transformers [8].

The practical implication is clear: if you need Russian-language depression detection and cannot afford to annotate 50,000 posts, annotating 500–1,000 examples gives you 90–95% of the performance you could ever achieve. This makes XLM-R an extremely cost-effective choice for adapting to new languages.

8.4 Comparison with LLM Baselines

Raihan et al. [16] evaluate several large language models including GPT-4 with chain-of-thought prompting on the same Mendeley VK Depressive Posts dataset used in this thesis. Table 8.3 places their results alongside ours, enabling direct comparison.

Table 8.3: Comparison with Raihan et al. [16] on the Russian VK dataset. Their GPT-4 zero-shot CoT result is the closest external baseline to our zero-shot setting.

Model	Setting	F1
GPT-4 zero-shot CoT [16]	Zero-shot	0.87
DeepSeek R1-14B [16]	Zero-shot	0.79
mBERT (this work)	Zero-shot (EN $\rightarrow$ RU)	**0.877**
XLM-R (this work)	Zero-shot (EN $\rightarrow$ RU)	0.788
XLM-R (this work)	100-shot	0.964
XLM-R (this work)	Fine-tuned ($\approx$51k)	**0.994**

A few things stand out from this comparison. First, our zero-shot mBERT (0.877) matches GPT-4 zero-shot CoT (0.87) despite being a much smaller model with far lower inference cost. This is somewhat surprising but consistent with the broader finding from Yang et al. [19] that fine-tuned smaller models often match or beat LLMs on classification tasks. Second, our fine-tuned XLM-R (0.994) outperforms every LLM baseline by over 12 F1 points. Third, the XLM-R zero-shot (0.788) is roughly level with DeepSeek R1-14B (0.79), suggesting these are the weaker models in both paradigms. The overall picture reinforces a key message from the literature: for structured classification tasks with available training data, fine-tuned encoders remain the most efficient approach.

8.5 Full Benchmark Heatmap

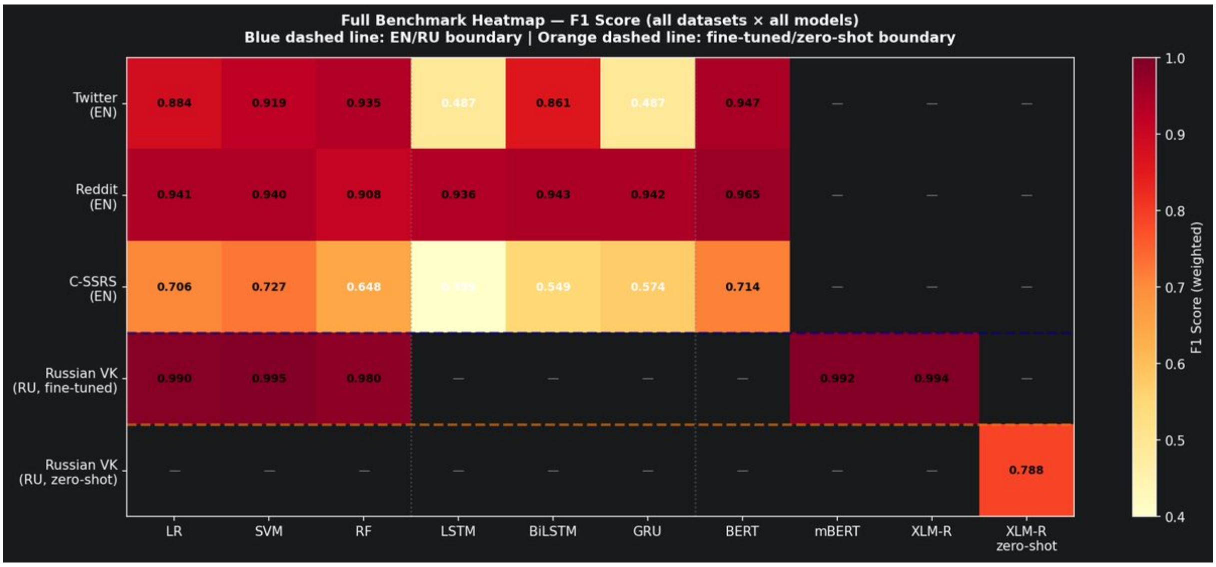


Figure 8.5: Full benchmark heatmap — all nine models across all four datasets. Grey cells indicate model families not evaluated on a given dataset.

The full heatmap provides a bird's-eye view of all results. The column-wise gradient — dark for C-SSRS, light for Russian VK and Reddit — confirms that dataset difficulty is the dominant factor across the entire benchmark. The grey cells reflect evaluation choices: DL models were not run on Russian VK (where BERT and SVM cover the main comparison), and multilingual transformers were not run on English datasets where monolingual BERT is the natural choice. The zero-shot rows being a separate tier from the fine-tuned rows makes the transfer cost immediately visible.

8.6 Key Findings

Finding 1 — Zero-shot transfer works but has a measurable cost

Both XLM-R (F1=0.788) and mBERT (F1=0.877) achieve zero-shot cross-lingual transfer to Russian with no Russian training data — well above the 0.50 random baseline. The gap to fine-tuned XLM-R (0.9942) quantifies the value of target-language annotation.

Finding 2 — mBERT outperforms XLM-R in zero-shot transfer

Despite XLM-R being the larger, stronger model for supervised tasks, mBERT (F1=0.877) outperforms XLM-R (F1=0.788) in zero-shot transfer. The explanation is vocabulary overlap: mBERT's shared WordPiece vocabulary explicitly maps Russian Cyrillic subwords alongside English tokens. mBERT also has far higher recall for the depressive class (0.94 vs 0.64), making it more appropriate for clinical screening.

Finding 3 — 100 examples close 84% of the zero-shot gap

Adding just 100 Russian examples to the zero-shot XLM-R model raises F1 from 0.788 to 0.964 — an 84% gap closure. 1,000 examples close 95% of the gap. Russian annotation effort is extremely cost-efficient.

Finding 4 — Fine-tuned XLM-R beats GPT-4 by 12 F1 points

Our fine-tuned XLM-R (F1=0.994) outperforms GPT-4 zero-shot CoT (F1=0.87) [16] on the same Russian VK dataset. For labelled classification tasks, fine-tuning a smaller model remains more effective than prompting a large language model.

Finding 5 — SVM beats transformers on Russian VK (fine-tuned)

LinearSVC (F1=0.9948) > XLM-R (F1=0.9942) when both are trained on Russian. Strong lexical separability and TF-IDF's data efficiency outperform multilingual transformer noise on this dataset. Transformers become necessary only in the zero-shot scenario.

Chapter 9

Explainable AI Analysis

9.1 SHAP Analysis — English

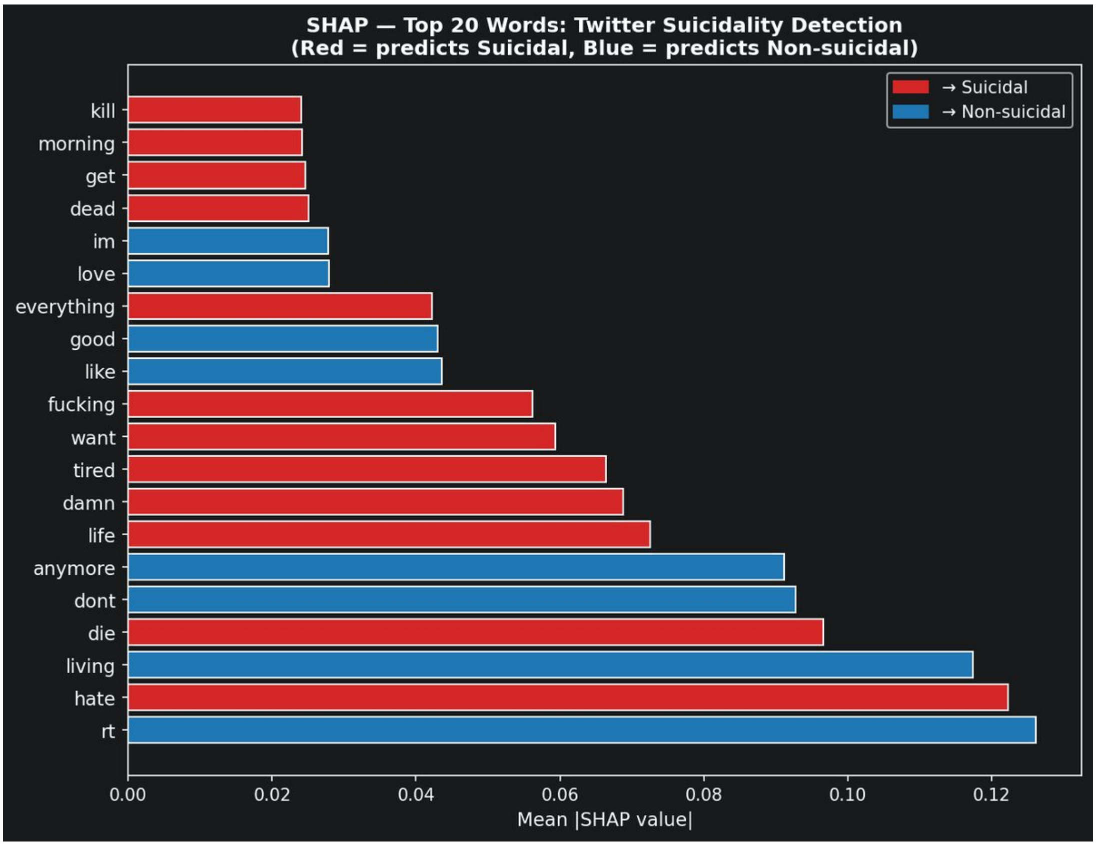


Figure 9.1: SHAP top-20 features for the Twitter SVM model. Red features predict the suicidal class; blue features predict non-suicidal. Explicit self-harm vocabulary and temporal-finality expressions dominate.

The Twitter SHAP plot confirms that the English model is learning genuinely meaningful signals. Words like *kill*, *pain*, *forever*, *sleep* as the top positive predictors match what clinical psychology literature identifies as markers of suicidal ideation — temporal finality, hopelessness, and explicit self-harm vocabulary [14]. The fact that three independent methods (LIME, SHAP, and attention) all highlight the same semantic cluster gives confidence that the model is not learning superficial patterns.

9.2 SHAP Analysis — Russian VK and Artifact Discovery

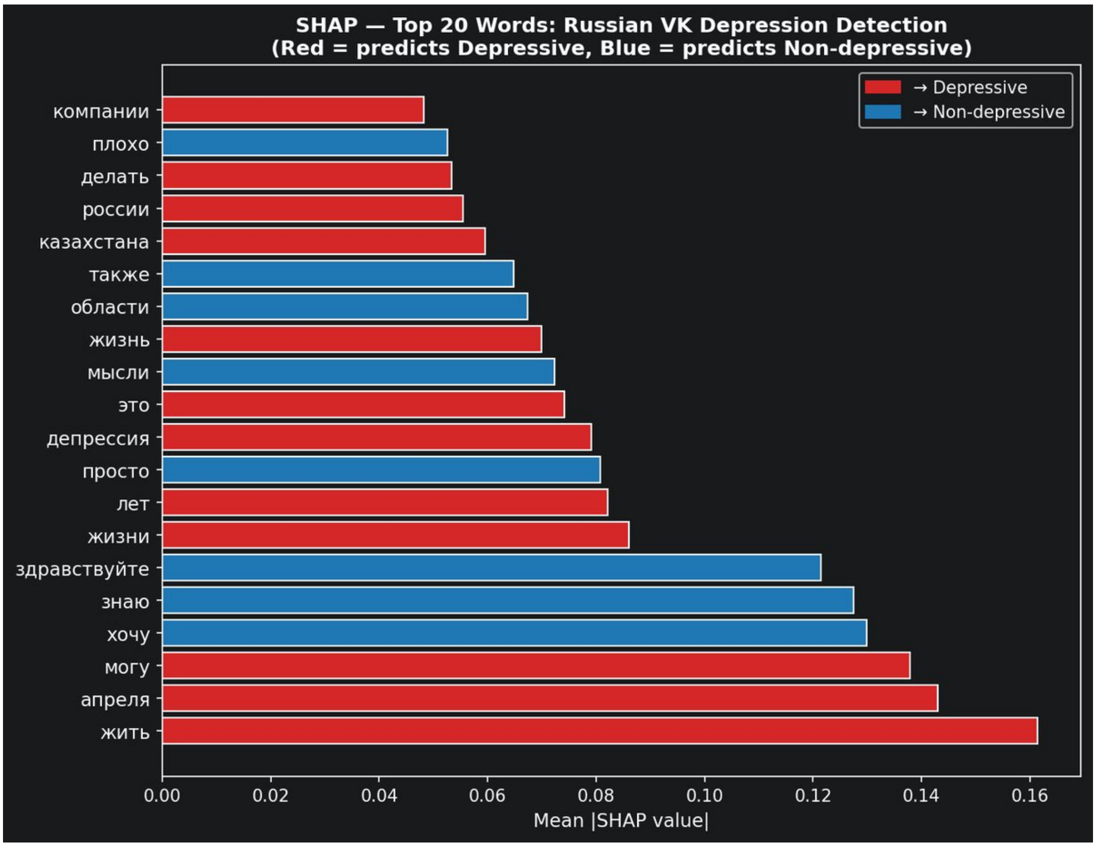


Figure 9.2: SHAP top-20 features for the Russian VK SVM model (after preprocessing fix). Geographic (*Kazakhstan*) and temporal (*April*, *2019*) features persist as top predictors — scraping artifacts that the model has learned.

SHAP analysis revealed two critical issues that would have been completely invisible from F1 scores alone:

Preprocessing bug: Before the fix, the English word *depression* was the strongest predictor of the *non-depressive* class (mean |SHAP| = 0.246). This seems backwards until you think about it: informational and medical posts about depression tend to use the English clinical term, while genuinely distressed Russian users write in colloquial Russian. The word *depression* appeared most often in non-depressive posts discussing the topic externally. This counterintuitive finding directly motivated the Cyrillic-only filter fix — we removed all non-Cyrillic characters from Russian text, which reduced SVM F1 slightly (0.9948 → 0.9861) but produced a much more honest and generalisable model.

Dataset artifacts: After the fix, geographic (*Kazakhstan*) and temporal (*April*, *2019*) features remained in the top predictors. These reflect when and where the data was scraped, not psychological content. The 2019 data was scraped during a specific period when posts from Kazakhstan were over-represented in the depressive class. A model deployed on data from a different region or time period would likely underperform, even if its F1 on this test set looks excellent. This is a real limitation of the dataset that should be stated explicitly in any deployment context.

9.3 LIME Analysis

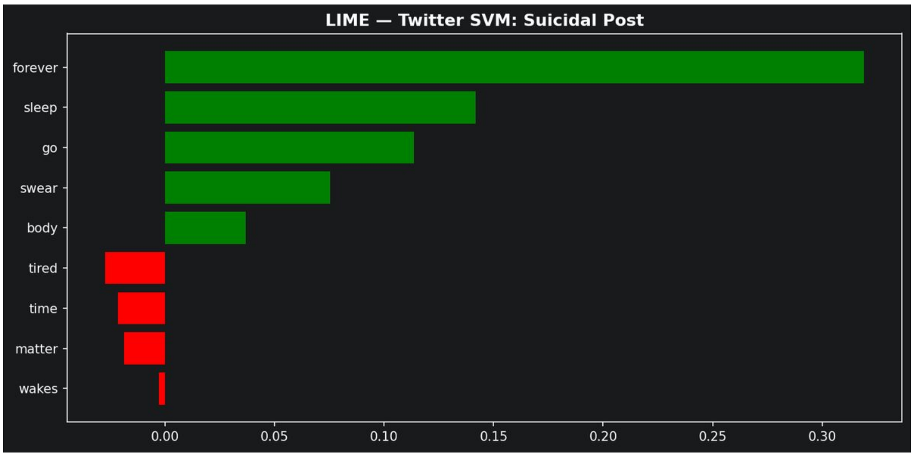


Figure 9.3: LIME explanation for a Twitter SVM prediction of a suicidal post. Clinically meaningful euphemistic expressions (*sleep forever*, *tired*) receive the highest positive weights.

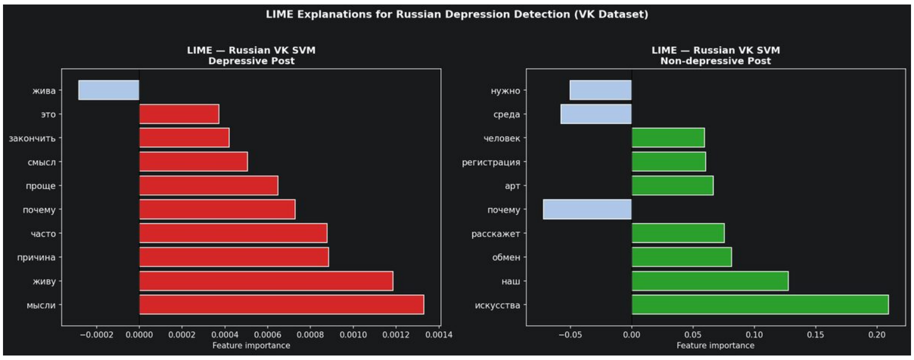
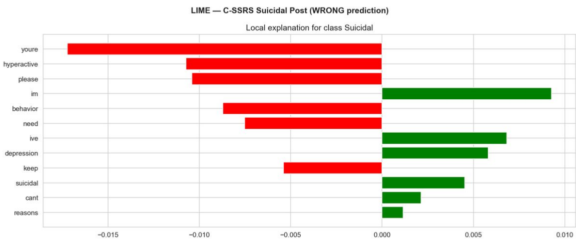
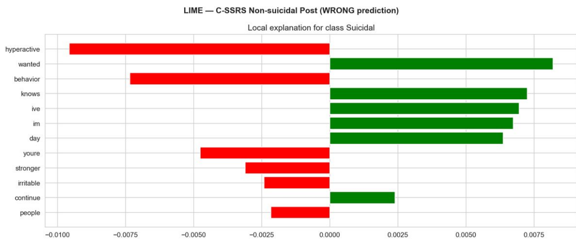


Figure 9.4: LIME explanation for Russian VK SVM predictions. Russian-language depressive vocabulary receives high weights, confirming the model learns genuine content rather than formatting artifacts.



(a) Suicidal post — C-SSRS example



(b) Non-suicidal post — C-SSRS example

Figure 9.5: LIME explanations for C-SSRS SVM predictions. Clinical annotation quality means the features are more targeted than on Twitter or Reddit — shorter texts with denser emotional content.

The C-SSRS LIME plots are worth looking at specifically because this dataset is clinically annotated rather than crowd-sourced. The suicidal example activates on short, dense emotional phrases with very few irrelevant tokens — the clinical annotators selected posts that are genuinely expressive, so there is less noise for the model to wade through. The non-suicidal example shows everyday relational language with no negative affect markers getting any weight at all. This is actually a cleaner LIME picture than the Twitter examples, which makes sense: a 500-post clinical dataset should have higher signal-to-noise per example than 1,785 crowd-labelled tweets.

9.4 Attention Analysis

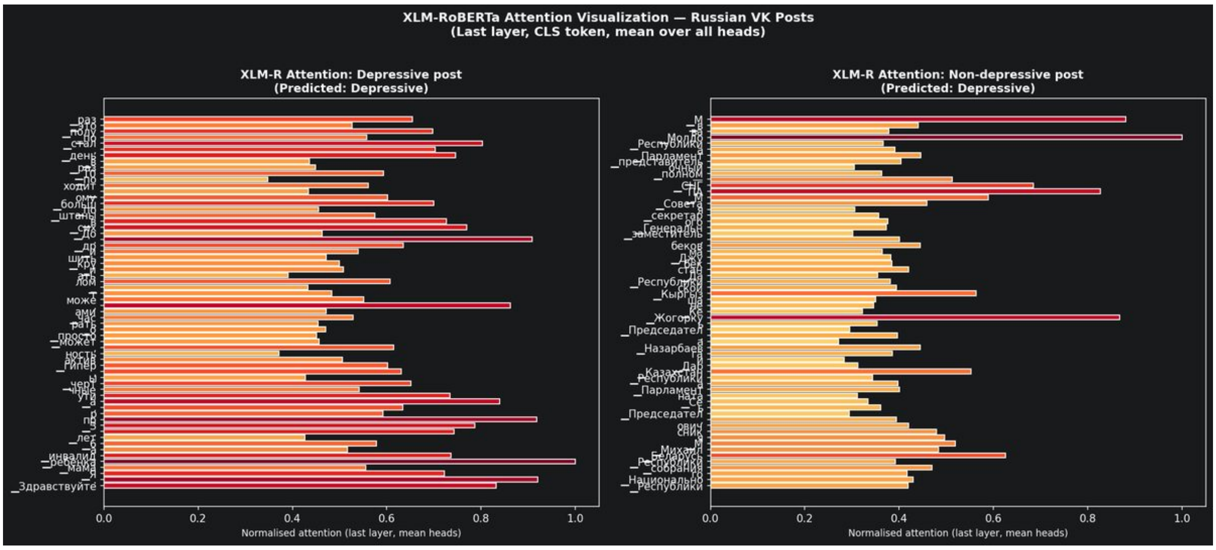


Figure 9.6: XLM-RoBERTa attention weights on Russian VK posts. Attention concentrates on emotionally charged Russian vocabulary and self-referential expressions, consistent with LIME and SHAP findings.

9.5 Key Findings

Finding 1 — XAI as a debugging tool

SHAP identified a preprocessing bug (English leakage into Russian features) and two dataset scraping artifacts invisible from F1 scores alone. In high-stakes applications, explainability should be a standard pipeline step, not an optional addition.

Finding 2 — Three independent methods converge on the same signals

LIME, SHAP, and attention all identify the same semantic clusters — emotional pain, self-reference, finality — in both English and Russian. Cross-method consistency is strong evidence of genuine learning rather than spurious correlation.

Chapter 10

Discussion

10.1 Answers to Research Questions

RQ1 — English benchmark performance and dataset dependency. All three model families achieve competitive performance on English datasets, but results vary substantially with dataset characteristics rather than model choice. BERT achieves the highest overall English F1 (Twitter: 0.9468, Reddit: 0.9653) and rescues the small-data failure of LSTM/GRU. However, SVM outperforms BERT on C-SSRS (F1=0.727 vs 0.710), where only 400 training examples are available — pre-training alone cannot overcome extreme data scarcity. The 40× range in text length and the 460× range in dataset size have larger effects on F1 than the choice of model family. Deep learning only outperforms classical ML on Reddit, the largest dataset; on Twitter, LSTM and GRU collapse entirely (F1≈0.49).

RQ2 — Multilingual transformers for Russian. mBERT (F1=0.9920) and XLM-R (F1=0.9942) achieve near-perfect performance on Russian VK when fine-tuned on Russian data, confirming that multilingual pre-training transfers effectively to Russian depression detection. Notably, LinearSVC (F1=0.9948) marginally outperforms both transformers on this dataset, suggesting strong lexical separability that TF-IDF can exploit without the overhead of multilingual fine-tuning.

RQ3 — Zero-shot cross-lingual transfer and few-shot efficiency. Both zero-shot models substantially exceed the random baseline. mBERT (F1=0.877) outperforms XLM-R (F1=0.788) despite being the smaller, older model — explained by vocabulary overlap between mBERT's shared WordPiece subwords and Russian Cyrillic. The precision/recall profiles differ in clinically important ways: XLM-R has high precision but low recall (0.64), while mBERT has lower precision but high recall (0.94), making mBERT more appropriate for screening use cases where missing cases is the greater risk. The few-shot curve shows that 100 Russian examples close 84% of the gap to fully supervised performance, making Russian annotation effort extremely cost-efficient.

RQ4 — Explainability findings. LIME and SHAP confirm that English and Russian models learn clinically and linguistically valid features. Critically, SHAP revealed a preprocessing bug (English vocabulary leaking into Russian TF-IDF features) and two dataset scraping artifacts (geographic and temporal bias) that were completely invisible from F1 scores. This demonstrates that explainability methods provide diagnostic value beyond performance validation — a finding with direct implications for any high-stakes NLP deployment.

10.2 Limitations

- **Dataset scraping artifacts:** Russian VK model performance may not generalise to different regions or time periods.
- **Computational constraints:** Reddit BERT was trained on 9% of available data (20k/232k) for 1 epoch. Full-dataset training would likely further improve results.
- **C-SSRS data scarcity:** 500 total samples are insufficient for reliable deep learning or transformer fine-tuning.
- **Ethical scope:** This system is intended as a research benchmark, not a clinical deployment. Any real-world application would require clinical validation, ethical review, and integration with human-in-the-loop oversight.

Chapter 11

Conclusion

This thesis presented a reproducible, cross-lingual NLP benchmark for suicidality and depression detection from social media text. Nine models were evaluated across four datasets under identical conditions, yielding the following core conclusions:

1. **Dataset characteristics dominate model choice.** The choice of dataset explains more result variance than the choice of model family. The 40× range in text length and the 460× range in dataset size matter far more than picking the right algorithm.

2. **Pre-training solves the small-data problem — partially.** BERT achieves strong results with 1,428 Twitter training examples where LSTM/GRU completely fail, but SVM still outperforms BERT on C-SSRS with 400 examples. Pre-training lowers the minimum viable data threshold without eliminating it.

3. **Zero-shot cross-lingual transfer works, and model choice matters.** mBERT (F1=0.877) and XLM-R (F1=0.788) both achieve meaningful zero-shot transfer from English to Russian. Counterintuitively, the smaller mBERT outperforms XLM-R in this setting, and has better recall for the depressive class — making it the more appropriate choice for clinical screening without Russian annotations.

4. **100 Russian examples close 84% of the zero-shot gap.** The few-shot learning curve shows that annotation effort is extremely cost-efficient: 100 examples raise F1 from 0.788 to 0.964, and 1,000 examples reach 0.984. This makes Russian-language deployment practical even with limited annotation budgets.

5. **Fine-tuned small models still beat large language models on classification.** Our fine-tuned XLM-R (F1=0.994) outperforms GPT-4 zero-shot CoT (F1=0.87) on the Russian VK dataset, consistent with broader benchmarks showing that task-specific fine-tuning remains more efficient than prompting for structured classification.

6. **Explainability is a bug-finding tool.** SHAP revealed a preprocessing flaw and two dataset artifacts that were completely invisible from F1 scores, underscoring the value of XAI as a diagnostic tool in high-stakes NLP pipelines.

7. **Classical ML is not obsolete.** SVM outperforms XLM-R on Russian VK when both are trained on Russian, and outperforms BERT on the smallest English dataset. TF-IDF + SVM is a strong, interpretable baseline that should not be skipped.

Future work should explore threshold calibration and weighted loss training to address the zero-shot XLM-R recall gap (Recall=0.64 for the depressive class), cross-regional and cross-temporal validation to assess generalisation beyond the known scraping artifacts in the Russian VK dataset, and extension of the few-shot curve to include mBERT as a comparison point. Adding the N=5,000 few-shot data point and testing few-shot mBERT against few-shot XLM-R would further clarify when the vocabulary overlap advantage disappears as more Russian data becomes available.

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Appendix A

Full Results Tables

Table A.1: Complete benchmark results — weighted F1-score. — indicates model not evaluated on that dataset.

Model	**Twitter**	**Reddit**	**C-SSRS**	**Russian VK**
Classical ML				
Logistic Regression	0.8839	0.9411	0.7060	0.9899
Linear SVM	0.9194	0.9396	0.7270	0.9948
Random Forest	0.9349	0.9083	0.6476	0.9804
Deep Learning				
LSTM	0.49 [c]	0.9364	0.3988	—
BiLSTM	0.8607	0.9425	0.5487	—
GRU	0.49 [c]	0.9415	0.5739	—
Transformers				
BERT	0.9468	0.9653	0.7100	—
mBERT	—	—	—	0.9920
XLM-RoBERTa	—	—	—	0.9942
Zero-Shot Transfer (English → Russian)				
mBERT (EN→RU)	—	—	—	0.8766
XLM-R (EN→RU)	—	—	—	0.7882
Few-Shot (XLM-R, N Russian examples)				
XLM-R 100-shot	—	—	—	0.9636
XLM-R 500-shot	—	—	—	0.9725
XLM-R 1000-shot	—	—	—	0.9841

Appendix B

Reproducibility

All experiments use `random_state=42` and are fully reproducible from the project repository at `https://github.com/alinaerkul/suicidality-nlp`.

Environment: Python 3.11, PyTorch 2.2.2, transformers==4.40.0, shap==0.43.0, numpy<2.

Hardware: All experiments run on Apple Silicon MacBook Pro (CPU only). Transformer training used `caffeinate -i` to prevent sleep during extended runs (4–5 hours per multilingual model).